Independence

Let $X_1, X_2 \sim$ continuous random variable with a joint PDF $f(x_1, x_2)$,

$$f(x_2|x_1) = \frac{f(x_1, x_2)}{f(x_1)}$$

$$\Rightarrow f(x_1, x_2) = f(x_2|x_1) \cdot f(x_1)$$

which is nothing more than a derivation of the multiplication rule: $P(A \cap B) = P(A|B) \cdot P(B)$ we saw in STAT101/231. Similar to the behaviour of independent probabilities, if $X_1, X_2$ are **independent RVs**,

$$f(x_2 \mid x_1) = f(x_2)$$

$$\Rightarrow f(x_1, x_2) = f(x_2) \cdot f(x_1)$$

and the equivalent for discrete independent random variables is

$$p(x_1, x_2) = p(x_1) \cdot p(x_2)$$

Definition 2.5.1: Let $X_1, X_2$ be random variables with some joint function, the random variables $X_1, X_2$ are **independent** $\Leftrightarrow f(x_1, x_2) = f(x_1) \cdot f(x_2) \equiv p(x_1, x_2) = p(x_1) \cdot p(x_2)$.

Theorem 2.5.1: Let $X_1, X_2$ be random variables with support $\mathcal{S}_1, \mathcal{S}_2$ with some joint function, then $X_1$ and $X_2$ are **independent** $\Leftrightarrow f(x_1, x_2)$ or $p(x_1, x_2)$ can be written as the production of non-negative functions

$$f(x_1, x_2) = g(x_1) \cdot h(x_2) \;\; \forall\, i \ni x_i \in \mathcal{S}_i$$

Example 2.13:

let $X_1, X_2 \sim$ cont. RV s.t. $f(x_1, x_2) = \begin{cases} 8x_1x_2, & 0 < x_1 < x_2 < 1 \\ 0, & e/w \end{cases}$

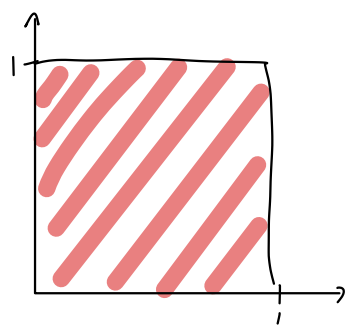
although $f(x_1, x_2)$ can be defined as the product of $g(x_1) \cdot h(x_2)$, there are issues with the support of $X_1, X_2$, i.e. if the bounds of $x_1$ is affected by $x_2$ (or vice versa), then clearly $X_1, X_2$ are not independent

Going back to example 2.2, we defined the joint PDF as:

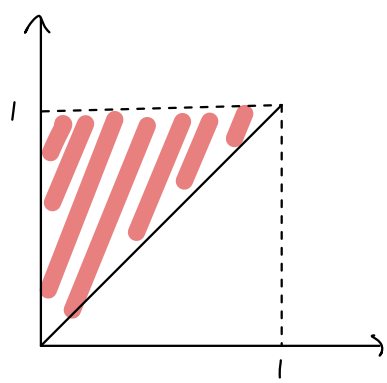
$$f(x_1, x_2) = \begin{cases} 6 \cdot x_1^2 \cdot x_2, & 0 < x_1, x_2 < 1 \\ 0, & \text{e/w} \end{cases}$$

And we can see a "separation" of the support, as well as the joint PDF as the product of two function, as described in Theorem 2.5.1:

$f(x_1, x_2) = 3x_1^2 \cdot 2x_2 = g(x_1) \cdot h(x_2)$ and the support can be veiwed as:




however, for example 2.13, where $0 < x_1 < x_2 < 1$, the support looks like




which highlights the bounds of $x_1$ is limited by $X_2$ and vice versa

Theorem 2.5.2: Let $(X_1, X_2)$ have a joint CDF $F(x_1, x_2)$ with marginal CDFs $F(x_1)$ and $F(x_2)$, then $X_1$ and $X_2$ are **independent** $\Leftrightarrow F(x_1, x_2) = F(x_1) \cdot F(x_2)\ \forall\ (x_1, x_2) \in \mathbb{R}^2$

Theorem 2.5.3: The random variables $X_1, X_2$ are **independent** if and only if the following condition holds true:

$$P(a < X_1 \leq b, c < X_2 \leq d) = P(a < X_1 \leq b) \cdot P(c < X_2 \leq d) \quad \forall\ a, b, c, d\ \in \mathbb{R}$$

It should be no surprise that Theorem 2.5.3 is derived from Theorem 2.5.2. Recall our proof for the calculating a joint probability interval (like above) based on the joint CDF:

$$\begin{aligned}
P(a < X_1 \leq b, c < X_2 \leq d) &= F(b,d) + F(a,c) - F(a,d) - F(b,c) \\
&= F(b)F(d) + F(a)F(c) - F(a)F(d) - F(b)F(c) \\
&= F(d)[F(b) - F(a)] + F(c)[F(a) - F(b)] \\
&= F(d)[F(b) - F(a)] - F(c)[F(b) - F(a)] \\
&= [F(b) - F(a)] \cdot [F(d) - F(c)] \\
&= Pr(a < X_1 < b) \cdot Pr(c < X_2 < d)
\end{aligned}$$

Example 2.14: Let the joint PDF of $X_1$ and $X_2$ be defined as:

$$f(x_1, x_2) = \begin{cases} x_1 + x_2, & 0 < x_1, x_2 < 1 \\ 0, & \text{e/w} \end{cases}$$

$$\begin{aligned}
Pr\left(0 < X_1 < \tfrac{1}{2}, 0 < X_2 < \tfrac{1}{2}\right) &= \int_0^{1/2}\int_0^{1/2} x_1 + x_2 \, dx_1 \, dx_2 \\
&= \ldots = \frac{1}{8}x_2 + \frac{1}{4}x_2^2 \Big|_0^{\frac{1}{2}} \\
&= \frac{1}{8}
\end{aligned}$$

If you look at the marginal, then $Pr(0 < X_1 < \frac{1}{2}) = \frac{3}{8}$ and $Pr(0 < X_2 < \frac{1}{2}) = \frac{3}{8}$ (symmetric in joint pdf)

Since $Pr(0 < X_1 < \frac{1}{2}, 0 < X_2 < \frac{1}{2}) = \frac{1}{8} \neq \frac{3}{8} \cdot \frac{3}{8} = Pr(0 < X_1 < \frac{1}{2}) \cdot Pr(0 < X_2 < \frac{1}{2})$

$X_1$ and $X_2$ are not independent of eachother

Not surprisingly, if $X_1, X_2$ are independent, then **any function of $X_1$ and $X_2$ are also independent**

Theorem 2.5.4: Suppose $X_1, X_2$ are independent and $E\big(u(X_1)\big)$ and $E\big(v(X_2)\big)$ $\exists$, then

$$E\big(u(X_1)\cdot v(X_2)\big) = E\big(u(X_1)\big)\cdot E\big(v(X_2)\big)$$

WLOG $E\big(u(x_1)\cdot v(x_2)\big) = \iint u(x_1)\cdot v(x_2)\cdot f(x_1,x_2)\, dx_1\, dx_2$

$= \iint u(x_1)\cdot v(x_2)\cdot f(x_1)\cdot f(x_2)\, dx_1\, dx_2$

$= \int v(x_2) f(x_2) \int u(x_1) f(x_1)\, dx_1\, dx_2$

$= \int v(x_2)\cdot f(x_2)\, dx_2 \cdot \int u(x_1) f(x_1)\, dx_1$

$= E\big(v(x_2)\big)\cdot E\big(u(x_1)\big)$

Theorem 2.5.5: Suppose the joint MGF $M(t_1, t_2)$ exists for the random variables $X_1, X_2$, then $X_1, X_2$ are independent if and only if $M(t_1, t_2) = M(t_1, 0)\cdot M(0, t_2)$

Example 2.15: Let the joint PMF of $X_1$ and $X_2$ be defined as:

$$p(x_1, x_2) = \left(\frac{1}{2}\right)^{x_1+x_2} \quad 1 \le x_1, x_2 < \infty$$

Since $p(x_1, x_2) = p(x_1)\cdot p(x_2)$ and the support has no "issues", then we should not be surprised that $X_1, X_2$ are independent.

However if we do have the theorems above, we can still prove that $X_1, X_2$ are independent via theorem 2.5.5

$M(t_1, t_2) = E\left(e^{t_1 x_1 + t_2 x_2}\right)$

$= \sum_{x_1}\sum_{x_2} e^{t_1 x_1 + t_2 x_2}\cdot\left(\frac{1}{2}\right)^{x_1+x_2}$

$= \sum_{x_1}\sum_{x_2} e^{t_1 x_1} e^{t_2 x_2}\left(\frac{1}{2}\right)^{x_1}\left(\frac{1}{2}\right)^{x_2}$

$= \sum_{x_1} e^{t_1 x_1}\left(\frac{1}{2}\right)^{x_1} \sum_{x_2} e^{t_2 x_2}\left(\frac{1}{2}\right)^{x_2}$

Via thm 2.5.5

$= E\left(e^{t_1 x_1}\right) E\left(e^{t_2 x_2}\right) \Rightarrow X_1, X_2$ ind

if we did not have thm 2.5.5, we recall

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \qquad \text{where } |r|<1$$

then let $r = \frac{1}{2}e^{t_i}$

$$\Rightarrow \sum_{x_1} e^{t_1 x_1}\left(\tfrac{1}{2}\right)^{x_1} \cdot \sum_{x_2} e^{t_2 x_2}\left(\tfrac{1}{2}\right)^{x_2}$$

$$= \sum_{x_1}\left(\tfrac{1}{2}e^{t_1}\right)^{x_1} \cdot \sum_{x_2}\left(\tfrac{1}{2}e^{t_2}\right)^{x_2}$$

and if we shift our $x$ to a starting point of 0

$$\therefore \sum_{x_1=1}^{\infty}\left(\tfrac{1}{2}e^{t_1}\right)^{x_1} = \frac{1}{1-r} - 1 = \frac{r}{1-r} = \frac{e^{t_1}}{2-e^{t_1}}$$

$$\sum_{x_2=1}^{\infty}\left(\tfrac{1}{2}e^{t_2}\right)^{x_2} = \left(\frac{e^{t_2}}{2-e^{t_2}}\right)$$

$$\Rightarrow \sum_{x_1}\sum_{x_2} e^{t_1 x_1 + t_2 x_2}\left(\tfrac{1}{2}\right)^{x_1+x_2} = \left(\frac{e^{t_1}}{2-e^{t_1}}\right)\left(\frac{e^{t_2}}{2-e^{t_2}}\right)$$

$$= M(t_1)\cdot M(t_2)$$

$$= M(t_1, t_2=0)\cdot M(t_1=0, t_2)$$

$$\Rightarrow X_1, X_2 \text{ independent}$$

Correlation

In the previous section, we talked about the idea of two random variables being independent of one another; however, in one of the previous examples with a joint PDF

$$f(x_1, x_2) = \begin{cases} x_1 + x_2, & 0 < x_1, x_2 < 1 \\ 0, & \text{e/w} \end{cases}$$

and we saw that $X_1, X_2$ are not independent of one another, but rather dependent. If these two variables dependent on one another, then there is a potential for $X_1$ to affect $X_2$, and vice-versa, which begs the question: how much influence does one have on the other?

Let $X$ and $Y$ be RVs and $E(X) = \mu_X$ and $E(Y) = \mu_Y$, then we can define the **covariance of $X$ and $Y$** as:

$$\begin{aligned} Cov(X, Y) &= E([X - \mu_X][Y - \mu_Y]) \\ &= E(X \cdot Y) - \mu_X \cdot \mu_Y \\ &= E(X \cdot Y) - E(X) \cdot E(Y) \end{aligned}$$

Look familiar? Recall that $\text{Var}(X) = E([X - \mu]^2) = E([X - \mu] \cdot [X - \mu])$, so the covariance is literally how the two random variables $X$ and $Y$ move together. If the greater values of one variable mainly correspond with the greater values of the other variable, i.e. if $(X - \mu_X)$ is large AND $(Y - \mu_Y)$ is large, and the same holds for smaller values, the covariance is positive. Not surprisingly, if one is large and the other small, the covariance is negative. This may sound familiar because, assuming $\sigma_X, \sigma_Y > 0$, then the **(Pearson's) correlation coefficient $\rho$** is defined as

$$\rho = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y} = \frac{Cov(X, Y)}{\sqrt{Var(X) \cdot Var(Y)}}$$

where $-1 \leq \rho \leq 1$. The correlation coefficient of $X$ and $Y$ gives us a measure (i.e. the strength) of the (linear) relationship between $X$ and $Y$.

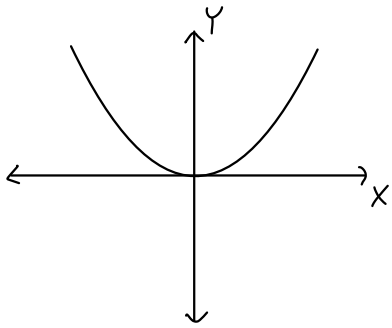
Although there is no rule on how we define the strength of an association, in general:

(1) $\rho > 0 \Rightarrow X, Y$ are positively ($\oplus$) associated
(2) $\rho < 0 \Rightarrow X, Y$ are negatively ($\ominus$) associated
(3) $\rho = 0 \Rightarrow$ no (linear) association

Two important takeaways from this are:

1) Since $\sigma_x, \sigma_y > 0$
$\Rightarrow$ sign of $\rho$ is a function/based on $Cov(x, y)$

2) $f(x) = X^2 = Y$




the following $\rho = 0$ for $X, Y$

* $\rho = 0$ does not imply no relationship/association, only that a linear relationship does not exist

Example 2.16:

$$f(x_1, x_2) = \begin{cases} x_1 + x_2, & 0 < x_1, x_2 < 1 \\ 0, & \text{e/w} \end{cases}$$

$$E(X_1) = \iint x_1 \cdot f(x_1, x_2)\, dx_2\, dx_1 = \frac{7}{12}$$

$$Var(x_1) = E(x_1^2) - E(x_1)^2 = \iint x_1^2 \cdot f(x_1, x_2)\, dx_2\, dx_1 - \left(\frac{7}{12}\right)^2$$

$$= \frac{11}{144}$$

$$\left.\begin{aligned} E(X_2) &= \frac{7}{12} \\ Var(X_2) &= \frac{11}{144} \end{aligned}\right\} f(x_1, x_2) \text{ is symmetric}$$

$$Cov(X_1, X_2) = E(X_1 X_2) - E(X_1) \cdot E(X_2)$$

$$= \int_0^1 \int_0^1 x_1 x_2 (x_1 + x_2)\, dx_1\, dx_2 - \left(\frac{7}{12}\right)\left(\frac{7}{12}\right)$$

$$= -\frac{1}{144}$$

$$\Rightarrow \rho = \frac{Cov(X_1, X_2)}{\sigma_{X_1} \cdot \sigma_{X_2}} = \frac{-\frac{1}{144}}{\sqrt{\frac{11}{144}} \cdot \sqrt{\frac{11}{144}}} = -\frac{1}{11} \approx -0.091$$

$< 0.5 \Rightarrow$ weak
$0.5 - 0.8 \Rightarrow$ moderate
$> 0.8 \Rightarrow$ strong

$\therefore X_1, X_2$ are "weakly" inversely related

It is important to note, though, that the correlation between two variables is a numerical relationship based on the idea that these two variables have a known marginal and joint distribution that they follow, or at least that we are willing to assume that the follow. This is important because $\rho$ is a population parameter, i.e. a known quantity, whereas $r$ is a sample statistic/estimate, something that is based on a sample that can be biased because of poor sampling, i.e. who and how we sample has led to biases. As such, **we must take extra caution when interpreting the numerical relationship between two variables based on $r$, particularly in the context of an association versus causation**.

Recall that $\rho$ relates to how much $X$ and $Y$ vary together, which could imply a relationship/association of some sort, which means there is some level of dependence. If $X$, $Y$ are independent, we know that

$$f(x_1, x_2) = f(x_1) \cdot f(x_2)$$

This property directly impacts our covariance:

$$\Rightarrow Cov(X_1, X_2) = E(X_1 X_2) - E(X_1)E(X_2)$$

$$= \iint x_1 x_2 f(x_1, x_2)\, dx_1\, dx_2 - E(X_1)E(X_2)$$

$$= \int x_1 f(x_1, x_2)\, dx_1 \cdot \int x_2 f(x_1, x_2)\, dx_2 - E(X_1)E(X_2)$$

$$= E(X_1)E(X_2) - E(X_1)E(X_2)$$

$$= 0$$

$\Rightarrow$ Assuming $\sigma_{x_1}, \sigma_{x_2}$ exist and are $>0$, $\rho = \frac{cov(x_1, x_2)}{\sigma_{x_1} \sigma_{x_2}} = 0$

if $X_1, X_2$ independent $\Rightarrow \rho = 0$, but if $\rho = 0 \not\Rightarrow X, Y$ indepentent

why? $\rho = 0$ implies no linear relationship

Let $f(x, y)$ be the joint PDF of $X, Y$, then

$$E(Y|X = x) = \int y \cdot f(y|x)\ dy \equiv \sum y \cdot p(y|x)$$

and if $Y = u(X)$, i.e. $y$ is a function of $X$, then if $u(X)$ is a linear function of $X \ni$

$$u(X) = a \cdot X + b$$

then $Y$ is a **linear conditional mean of $\boldsymbol{X}$**.

Question: Does this look familiar?

$A = u(X) = aX + b = \beta_0 + \beta_1 x$, ie., a linear model

Theorem 2.4.1: Suppose $(X, Y)$ have a joint distribution $\ni \exists\ \sigma_X^2, \sigma_Y^2 > 0$, then $\rho$ is the correlation coefficient of $X, Y$ and $E(Y|X)$ is a linear function then

$$E(Y|X) = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(X - \mu_X)$$

and

$$E(Var(Y|X)) = \sigma_Y^2(1 - \rho^2)$$

$$E(Y|X) = \mu_y + \rho\frac{\sigma_y}{\sigma_x}(x - \mu_x)$$

produces the simple linear model

$$y = \beta_0 + \beta_1 x = mx + b$$
$$\approx \hat{\beta}_0 + \hat{\beta}_1 x$$
$$= \bar{y} + \hat{\beta}_1(x - \bar{x})$$

$$\Rightarrow \beta_1 = \rho\frac{\sigma_y}{\sigma_x} = \frac{Cov(x,y)}{\sigma_x \cdot \sigma_y} \cdot \frac{\sigma_y}{\sigma_x} = \frac{Cov(x,y)}{\sigma_x^2}$$

recall STAT 231: $\hat{\beta}_1 = \dfrac{\sum x_i y_i - \frac{\sum x_i y_i}{n}}{\sum (x_i - y_i)^2} = \dfrac{S_{xy}^2}{S_x^2} = \dfrac{Cov(x,y)}{\sigma_x^2}$

$$\Rightarrow r = \frac{S_{xy}^2}{S_x \cdot S_y}$$

Example 2.17: Let $X$ and $Y$ be random variables $\ni \exists\ E(Y|X)$ and $E(X|Y)$ defined as:

$E(Y|X) = 4x + 3$ $\qquad$ $E(X|Y) = \frac{1}{16}y - 3$

$$= \mu_y + \rho\frac{\sigma_y}{\sigma_x}(x - \mu_x)$$

In the special case, where $X = \mu_X$ and $Y = \mu_Y$, we can solve specifically for $\mu_X$ and $\mu_Y$:

if $X = \mu_x \Rightarrow E(Y|X) = 4x + 3 = 4\mu_x + 3$

recall: $E(E(X|Y)) = E(x) = \mu_x$

$\Rightarrow E(E(X|Y)) = E\left(\frac{1}{16}Y - 3\right) = \frac{1}{16}\mu_y - 3 = \mu_x$

$\Rightarrow 4\mu_x + 3 = 4\left(\frac{1}{16}\mu_y - 3\right) + 3$

if $Y = \mu_y \Rightarrow E(X|Y) = \mu_x = \frac{1}{16}\mu_y - 3$ and so using a similar argument as above,

$E(E(Y|X)) = \mu_y = 4\mu_x + 3$

$\Rightarrow \frac{1}{16}\mu_y - 3 = \frac{1}{16}(4\mu_x + 3) - 3$

solving for this special case where,

$\mu_x = \frac{1}{4}\mu_x - \frac{45}{16} \Rightarrow \frac{3}{4}\mu_x = -\frac{45}{16}$

$\Rightarrow \mu_x = -\frac{15}{4}$

$\mu_y = \frac{1}{4}\mu_y - 9 \Rightarrow \mu_y = -12$

Rearranging our previous example to look more like a linear function:

$$E(Y|X = x) = 3 + 4x = \left(\mu_y - \mu_x \cdot \rho \frac{\sigma_y}{\sigma_x}\right) + \rho\left(\frac{\sigma_y}{\sigma_x}\right) \cdot x$$

$$E(X|Y = y) = -3 + \frac{1}{16}y = \left(\mu_x - \mu_y \cdot \rho \frac{\sigma_x}{\sigma_y}\right) + \rho\left(\frac{\sigma_x}{\sigma_y}\right) \cdot y$$

if we multiply the coefficients

$$\rho^2\left(\frac{\sigma_x}{\sigma_y}\right)\left(\frac{\sigma_y}{\sigma_x}\right) = 4\left(\frac{1}{16}\right) = \frac{1}{4} \Rightarrow \rho = \frac{1}{2} = 0.5$$

$\beta_1$ from $E(Y|X)$

$\beta_1$ from $E(X|Y)$

Under this special case, we can solve for $\mu_X, \mu_Y, \rho$, and the ratio $\sigma_Y/\sigma_x$ and $\sigma_X/\sigma_Y$, where the latter is obtained by taking the quotient of the coefficients

$$\frac{\rho\left(\frac{\sigma_y}{\sigma_x}\right)}{\rho\left(\frac{\sigma_x}{\sigma_y}\right)} = \frac{\sigma_y^2}{\sigma_x^2} = \frac{4}{1/16} = 64 \quad \Rightarrow \frac{\sigma_y}{\sigma_x} = 8, \ \frac{\sigma_x}{\sigma_y} = \frac{1}{8}$$

Question: Although $\rho^2 = 1/4 \ \Rightarrow \rho = \pm 1/2$, we know that $\rho > 0$. Why?

Given $\rho = \dfrac{Cov(X,Y)}{\sigma_x \cdot \sigma_y}$ and $\beta_1 = \rho \cdot \dfrac{\sigma_y}{\sigma_x}$

since $\beta_1 > 0 \ \forall \ E(X|Y), E(Y,X)$

$\Rightarrow \rho > 0$

$\hookrightarrow 3 + 4x$ $\quad$ $\hookrightarrow -3 + \frac{1}{16}y$

Recall that some important properties of the mean and variance are:

(1) $E(aX + b) = a\,E(X) + b,\ \forall\, a, b \in \mathbb{R}$
(2) $Var(aX + b) = a^2\, Var(X)\ \ \forall\, a, b \in \mathbb{R}$

As such, some important properties of correlation are:

$$Corr(aX+b,\ cY+d) = Corr(X, Y) \quad \forall\, a,b,c,d \in \mathbb{R}$$

$$\Rightarrow Corr(aX+b,\ Y) = Corr(X, Y)$$

$$\Rightarrow Corr(mX+b,\ Y) = Corr(X, Y)$$

$$Corr(X, Y) = \frac{Cov(X, Y)}{\sigma_x\, \sigma_y}$$

try for yourself to prove

$$Corr(aX+b,\ cY+d) = Corr(X, Y)$$

$$\hookrightarrow \frac{a \cdot c \cdot Cov(X, Y)}{\sqrt{a^2 Var(X) \cdot c^2 Var(Y)}}$$

Example 2.18: Let $X, Y \sim$ discrete random variables with joint PMF defined by:

| | | $X$ | | |
|---|---|---|---|---|
| | | 1 | 2 | $P_Y(y)$ |
| $Y$ | 1 | $\frac{2}{15}$ | $\frac{1}{15}$ | $\frac{3}{15}$ |
| | 2 | $\frac{4}{15}$ | $\frac{1}{15}$ | $\frac{5}{15}$ |
| | 3 | $\frac{3}{15}$ | $\frac{4}{15}$ | $\frac{7}{15}$ |
| | $P_X(x)$ | $9/15$ | $6/15$ | |